Statistical Testing of Pseudo-random Sequences in Monte Carlo Simulation

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February 2026

1 Executive Summary

In this project, we discuss and implement several statistical tests used to assess the quality of pseudo-random number generators. The objective is to determine whether a sequence of generated values behave as independent samples from the Uniform ($U(0,1)$) distribution. We present the underlying methodology for each test, derive the associated test statistics, and apply them to sequences produced by selected generators. The empirical results are presented and analyzed to validate the reliability of these Monte Carlo estimators.

2 Objectives

We investigate statistical testing based on two goodness-of-fit tests: the χ^2-test and the Kolmogorov-Smirnov (KS) test. For each method, we define the test statistic and discuss the implication of the test results. Specifically, we aim to do the following:

- Prove a simplification of the KS-statistic.
- Compute a numerical example of the KS-Statistic.
- Implement the KS test to a sequence generated by a Fibonacci generator.
- Discuss and provide a numerical example of the Serial Correlation Test.
- Describe the run-up test and apply it to a Multiplicative Congruential Generator (MCG).

All algorithms are constructed in the Julia programming language (v.1.12.3). We provide relevant numerical and visual results. The complete code is included in the Appendix [5]. Specific problems are stated in "Problem boxes," and are found in [1].

3 Pseudo-random Sequences

Pseudorandom Number Generators (PRNGs) are deterministic algorithms that produce sequences of real numbers whose statistical properties are examined through finitely many empirical tests. PRNGs are designed to mimic the behavior of independent random samples. We will focus on generating pseudo-random numbers from $U(0,1)$ (the Uniform distribution on $(0,1)$).
A **linear congruential generator** (LCG) is a PRNG defined by the recurrence relation:

$$x_n \equiv a x_{n-1} + c \mod M, \tag{1}$$

where M is the modulus, $a, c \in \{0, 1, \ldots, M-1\}$ are the multiplier and increment, respectively, and x_0 is the seed of the generator. The LCG produces a periodic sequence of at most M distinct integers $x_1, x_2, \ldots$. To obtain pseudo-random numbers between 0 and 1, we take

$$u_n = \frac{x_n}{M}. \tag{2}$$

If $c = 0$ then the LCG is called a **multiplicative congruential generator** (MCG), such that

$$x_n \equiv a x_{n-1} \mod M, \tag{3}$$

which can produce a sequence of at most $M-1$ distinct integers. Congruential generators may also have dependence on more than one term in the recursion. For example, we will consider the **lagged Fibonacci generator**, defined as

$$x_n \equiv x_{n-r} + x_{n-s} \mod M \tag{4}$$

for $n \geq r, s \in (0, r)$, where r and s are called the *lags* of the generator. We will evaluate the quality of the above generators by applying statistical tests to the sequences of pseudorandom numbers obtained from the generators.

4 Statistical Tests

To assess the effectiveness of a PRNG, we introduce statistical procedures used to determine whether the pseudorandom sequences behave as independent random samples from $U(0,1)$.

4.1 Goodness-of-Fit Tests

4.1.1 Chi-Squared Test

The χ^2 **goodness-of-fit test** is used to determine if an observed frequency distribution of a sample of categorical data matches the data's theoretical distribution. Suppose we have k mutually exclusive outcomes $A_1, \ldots, A_k$ each with

probablity $p_i = P(A_i)$, $i = 1, 2, \ldots, k$ of occurring. After performing n trials (or generating n pseudorandom values), let Y_i denote the number of observations falling in outcome A_i. Under the model, the *expected* number of observations in each category is np_i. The test is performed by calculating the χ^2-statistic:

$$Q_{k-1} = \sum_{i=1}^{k} \frac{(Y_i - np_i)^2}{np_i}. \tag{5}$$

Under the null hypothesis,

$$H_0: \text{ the sample follows the specified distribution,}$$

the statistic Q_{k-1} is approximately distributed as $Q_{k-1} \sim \chi^2(k-1)$, if the the expected counts np_i are sufficiently large (rule of thumb: $np_i \geq 5$).

4.1.2 Kolmogorov-Smirnov Test

For continuous data, we use the **Kolmogorov-Smirnov (KS) test** to compare the empirical distribution of a sample to a specified theoretical distribution. Let $X_1, X_2, \ldots, X_N$ be random samples with continuous distribution function $F(x)$. We can calculate the KS-statistic by taking

$$D_N = \sup_{-\infty < x < \infty} |F_N(x) - F(x)| \tag{6}$$

where $F_N(x)$ is the empirical (sample) distribution function of the observed data. If the $F(x)$ is completely determined, then the distribution of the KS-statistic is independent of $F(x)$, and we can use the following critical values for chosen significance level α:

10%	5%	1%
1.224	1.358	1.628

Table 1: Significance levels for $F(x)$

Under the null hypothesis,

$$H_0: \text{ sample follows distribution function F(x).}$$

When D_N is small ($< \alpha$), we fail to reject H_0 and can conclude the empirical distribution mimics the theoretical distribution.
Note that equation [6] can be expanded. Suppose we sort our sample data such that $X_1 \leq X_2 \leq \cdots \leq X_N$. Then

$$F_N(x) = \begin{cases} 0 & \text{if} \quad x < X_1 \\ \frac{k}{N} & \text{if} \quad X_k \leq x < X_{k+1} \\ 1 & \text{if} \quad x > X_1 \end{cases} \tag{7}$$

and the KS-statistic becomes

$$D_N = \max\left[\sup_{x<X_1} |-F(x)|, \sup_{X_1<x<X_2}\left|\frac{1}{N} - F(x)\right|, \ldots, \sup_{x>X_N} |1-F(x)|\right]. \quad (8)$$

The first and last terms simplify to

$$\sup_{x<X_1} |-F(x)| = F(X_1),$$
$$\sup_{x>X_N} |1-F(x)| = 1 - F(X_N).$$

We show the following:

Chapter 2 Problem 1

Prove

$$\sup_{x_k \le x < x_{k+1}} \left|\frac{k}{N} - F(x)\right| = \max\left(F(x_{x_{k+1}}) - \frac{k}{N}, \frac{k}{N} - F(x_k)\right) \quad (9)$$

for $k = 1, \ldots, N-1$.

Proof. Let $F_N(x)$ be defined as in [7] such that $F_N(x) = \frac{k}{N}$ if $x \in [x_k, x_{k+1})$.
Case 1: Show

$$\sup_{x_k \le x < x_{k+1}} \left|\frac{k}{N} - F(x)\right| \le \max\left(F(x_{x_{k+1}}) - \frac{k}{N}, \frac{k}{N} - F(x_k)\right).$$

By definition, $F(x)$ is non-decreasing, hence for any $x \in [x_k, x_{k+1})$ we have

$$F(x_k) \le F(x) \le F(x_{k+1}).$$

Now, observe that the absolute value $|\frac{k}{N} - F(x)| \in \mathbb{R}$ is a convex function. Therefore it has a maximum at either endpoint on interval $[x_k, x_{k+1}]$, such that for every $x \in [x_k, x_{k+1})$,

$$\left|\frac{k}{N} - F(x)\right| \le \max\left(\left|\frac{k}{N} - F(x_k)\right|, \left|\frac{k}{N} - F(x_{k+1})\right|\right).$$

Hence taking the supremum over $x \in [x_k, x_{k+1})$ still gives

$$\sup_{x_k \le x < x_{k+1}} \left|\frac{k}{N} - F(x)\right| \le \max\left(\left|\frac{k}{N} - F(x_k)\right|, \left|\frac{k}{N} - F(x_{k+1})\right|\right).$$

Case 2: Show

$$\sup_{x_k \le x < x_{k+1}} \left|\frac{k}{N} - F(x)\right| \ge \max\left(F(x_{x_{k+1}}) - \frac{k}{N}, \frac{k}{N} - F(x_k)\right).$$

Since $x > x_k$ on the interval, we have

$$\sup_{x_k \le x < x_{k+1}} \left| \frac{k}{N} - F(x) \right| \ge \left| \frac{k}{N} - F(x_k) \right|.$$

Furthermore, by the continuity of $F(x)$, for any $\varepsilon > 0$ there exists $x \in [x_k, x_{k+1})$ such that

$$|F(x) - F(x_{k+1})| < \varepsilon.$$

Following from the above inequality,

$$\left| F(x) - \frac{k}{N} + \frac{k}{N} - F(x_{k+1}) \right| < \varepsilon \quad \Longrightarrow \quad \left| \left(\frac{k}{N} - F(x_{k+1}) \right) - \left(\frac{k}{N} - F(x) \right) \right| < \varepsilon$$

Applying the reverse triangle inequality,

$$\begin{aligned}
&\left| \left| \frac{k}{N} - F(x_{k+1}) \right| - \left| \frac{k}{N} - F(x) \right| \right| \le \left| \left(\frac{k}{N} - F(x_{k+1}) \right) - \left(\frac{k}{N} - F(x) \right) \right| < \varepsilon \\
&\Longrightarrow -\varepsilon < \left| \frac{k}{N} - F(x_{k+1}) \right| - \left| \frac{k}{N} - F(x) \right| < \varepsilon \\
&\Longrightarrow \left| \frac{k}{N} - F(x_{k+1}) \right| < \left| \frac{k}{N} - F(x) \right| + \varepsilon \\
&\Longrightarrow -\varepsilon + \left| \frac{k}{N} - F(x_{k+1}) \right| < \left| \frac{k}{N} - F(x) \right|
\end{aligned}$$

Taking $\varepsilon \to 0$ gives us

$$\sup_{x_{(k)} \le x < x_{(k+1)}} \left| \frac{k}{N} - F(x) \right| \ge \left| \frac{k}{N} - F(x_{k+1}) \right|.$$

Combining with the previous result yields

$$\sup_{x_{(k)} \le x < x_{(k+1)}} \left| \frac{k}{N} - F(x) \right| \ge \max\left(\left| \frac{k}{N} - F(x_k) \right|, \left| \frac{k}{N} - F(x_{k+1}) \right| \right).$$

Lastly, since $F_N(x) = \frac{k}{N}$ for $x \in [x_k, x_{k+1})$, we can remove the absolute values, where

$$\left| \frac{k}{N} - F(x_k) \right| = \frac{k}{N} - F(x_k) \quad \text{and} \quad \left| \frac{k}{N} - F(x_{k+1}) \right| = F(x_{k+1}) - \frac{k}{N}.$$

Hence we achieve the desired equation:

$$\sup_{x_k \le x < x_{k+1}} \left| \frac{k}{N} - F(x) \right| = \max\left(F(x_{x_{k+1}}) - \frac{k}{N}, \frac{k}{N} - F(x_k) \right).$$

$\square$

The KS-statistic therefore simplifies to

$$\begin{aligned} D_N &= \max\left[F(X_1), \max_{k=1,\dots,N-1}\left(F(x_{x_{k+1}}) - \frac{k}{N}, \frac{k}{N} - F(x_k)\right), 1 - F(X_N)\right] \\ &= \max\left[\max_{k=1,\dots,N}\left(\frac{k}{N} - F(x_k)\right), \max_{k=1,\dots,N}\left(F(x_k) - \frac{k-1}{N}\right)\right], \end{aligned}$$

which can be defined by the shorthand notation

$$D_N = \max(D^+, D^-). \tag{10}$$

In practice, we may also compute the modified KS-statistic:

$$\tilde{D}_N = D_N(\sqrt{N} + 0.12 + 0.11/\sqrt{N}). \tag{11}$$

The significance values for $\tilde{D}_N$ remain the same as in table [1].

4.2 Empirical Tests

In this section, we define and describe two statistical tests for PRNGs. In the following section, we implement the tests on pseudorandom sequences and present the results.

4.2.1 Serial Correlation Test

The **serial correlation test** checks whether a sample $u_1, u_2, \dots, u_N$ mimics an i.i.d. sequence of uniform random variables (this is our null hypothesis). We generate the pseudorandom sequence and define two samples:

$$\begin{aligned} U &: u_1, u_2, \dots, u_{N-1}, u_N, \\ V &: u_2, u_3, \dots, u_N, u_1. \end{aligned}$$

Then, we compute the correlation coefficient of the two samples, such that

$$\rho = \frac{N\sum_{i=1}^{N} u_i u_{i+1} - (\sum_{i=1}^{N} u_i)^2}{N\sum_{i=1}^{N} u_i^2 - (\sum_{i=1}^{N} u_i)^2} \tag{12}$$

for $u_{N+1} = u_1$. Recall ρ takes values between -1 and 1, indicating the degree of correlation between the pairs of random variables. We expect ρ to fall in the 95% confidence interval $(\mu_N - 2\sigma_N, \mu_N + 2\sigma_N)$, where

$$\mu_N = -\frac{1}{N-1} \quad \text{and} \quad \sigma_N^2 = \frac{N^2}{(N-1)^2(N-2)}. \tag{13}$$

If ρ is large (does not belong to the confidence interval with 95% probability), then we reject the null hypothesis and claim dependence between successive pairs of the sequence.

4.2.2 Run Test

Suppose we have a sequence $x_1, x_2, \ldots, x_N$ of pseudo-random numbers. The **run test** checks whether the sample is independent by observing the arrangement of the values in the sequence. For a **run-up** test, we define the following event: a "run-up of length i" is any subsequence $\{x_j, \ldots, x_{j+i}\}$ such that

$$x_j \leq x_{j+1} \leq \cdots \leq x_{j+i-1} \quad \text{and} \quad x_{j+i-1} \geq x_{j+i}. \tag{14}$$

We fix the maximum run-up length to be k, and let event "run-up of length k" include run-ups of length k or more. Denote $u(i)$ to be the count of run-ups of length i, for $i = 1, 2, \ldots, k-1$, and $u(k)$ to be the count of run-ups of length k or more. We calculate the run-up statistic:

$$r_u = \frac{1}{N} \sum_{i=1}^{k} \sum_{j=1}^{k} (u(i) - Nb_i)(u(j) - Nb_j) a_{i,j}, \tag{15}$$

where a and b are specified coefficients. For $k = 6$, the coefficients are

$$b = \left[\frac{1}{6}, \frac{5}{24}, \frac{11}{120}, \frac{19}{720}, \frac{29}{540}, \frac{1}{840} \right], \tag{16}$$

$$a = \begin{bmatrix} 4529.4 & 9044.9 & 13568 & 18091 & 22615 & 27892 \\ 9044.9 & 18097 & 27139 & 36187 & 45234 & 55789 \\ 13568 & 27139 & 40721 & 54281 & 67852 & 83685 \\ 18091 & 36187 & 54281 & 72414 & 90470 & 111580 \\ 22615 & 45234 & 67852 & 90470 & 113262 & 139476 \\ 27892 & 55789 & 83685 & 111580 & 139476 & 172860 \end{bmatrix}. \tag{17}$$

When the pseudo-random sequence is large ($n \geq 4000$), then $r_u \sim \chi^2(k)$. If r_u is too large or too small, meaning we have too many or too few monotonically increasing subsequences, then we *reject* the null hypothesis of independence.
The construction of a **run-down** test follows easily. We define a "run-down of length i" to be any subsequence $\{x_j, \ldots, x_{j+i}\}$ such that

$$x_j \geq x_{j+1} \geq \cdots \geq x_{j+i-1} \quad \text{and} \quad x_{j+i-1} \leq x_{j+i}. \tag{18}$$

We calculate the run-down statistic r_d by replacing $u(\cdot)$ in [15] with $d(\cdot)$, where $d(i)$ is the number of run-down of length i, for $i = 1, 2, \ldots, k-1$, and $d(k)$ is the number of run-down of length k or more. The coefficients remain the same for both tests when $k = 6$.

5 Description and Analysis of Results

We first provide an example for computing the KS statistic [6].

Chapter 2 Problem 2

Compute $D_N = \max(D^+, D^-)$ for the data set $x_1 = 0.2$, $x_2 = 0.6$, $x_3 = 0.7$. Take F to be the CDF of $U(0,1)$; the uniform distribution on $(0,1)$. What do you think D^+, D^-, and D_N measure, intuitively?

We have the following cdf [7]:

$$F(x) = \begin{cases} 0 & \text{if} \quad x < 0 \\ x & \text{if} \quad 0 \leq x < 1 \\ 1 & \text{if} \quad x \geq 1 \end{cases}$$

We compute D^+ and D^- [10] for $N = 3$, where

$$\begin{aligned} D^+ &= \max_{k=1,\ldots,3} \left(\frac{k}{N} - F(x_k) \right) \\ &= \max \left(\frac{1}{3} - F(x_1), \frac{2}{3} - F(x_2), 1 - F(x_3) \right) \\ &= \max \left(\frac{1}{3} - F(0.2), \frac{2}{3} - F(0.6), 1 - F(0.7) \right) \\ &= \max \left(\frac{1}{3} - 0.2, \frac{2}{3} - 0.6, 1 - 0.7 \right) \\ &= \max \left(\frac{2}{15}, \frac{1}{15}, \frac{3}{10} \right) \\ &= \frac{3}{10} = 0.3 \end{aligned}$$

and

$$\begin{aligned} D^- &= \max_{k=1,\ldots,3} \left(F(x_k) - \frac{k-1}{N} \right) \\ &= \max \left(F(x_1), F(x_2) - \frac{1}{3}, F(x_3) - \frac{2}{3} \right) \\ &= \max \left(F(0.2), F(0.6) - \frac{1}{3}, F(0.7) - \frac{2}{3} \right) \\ &= \max \left(0.2, 0.6 - \frac{1}{3}, 0.7 - \frac{2}{3} \right) \\ &= \max \left(\frac{1}{5}, \frac{4}{15}, \frac{1}{30} \right) \\ &= \frac{4}{15} \approx 0.267. \end{aligned}$$

Therefore

$$D_3 = \max(0.3, 0.267) = 0.3.$$

Intuitively, D^+ gives us the maximum possible right-sided difference between $F_N(x)$ and $F(x)$), capturing observations *larger* than what we expect for a sample from $U(0,1)$. Similarly, D^- gives us the maximum possible left-sided difference between $F(x)$) and $F_N(x)$, capturing observations *smaller* than what we expect for a sample from $U(0,1)$. Taking the maximum of D^+ and D^- tells us how close (or how far) the empirical distribution of our sample $\{0.2, 0.6, 0.7\}$ is to Uniform distribution on $(0,1)$.
Now, we present a numerical application of the Kolmogorov-Smirnov test, where we examine the performance of a Fibonacci generator [4]:

Chapter 2 Problem 4

Consider the Fibonacci generator: $x_{n+1} \equiv x_n + x_{n-1} \mod 2^{31}$, where $x_0 = x_1 = 1$. Apply the Kolmogorov-Smirnov test to the first 1000 numbers (including the seeds) obtained from this generator. What are your conclusions?

We obtain the first 1000 numbers from the Fibonacci generator:

```
[1, 1, 2, 3, 5, ... 585622089, 1677835952, 115974393, 1793810345,
↪  1909784738, 1556111435]
```

Transforming the sample to the Uniform distribution [2], the sequence becomes

```
[4.656612873077393e-10, 4.656612873077393e-10, 9.313225746154785e-10,
↪  1.3969838619232178e-9, 2.3283064365386963e-9, ...
↪  0.27270153583958745, 0.7813032492995262, 0.054004785139113665,
↪  0.8353080344386399, 0.8893128195777535, 0.7246208540163934]
```

We want to apply the KS test to the generated sample above. We examine the results under the following null hypothesis:

$$H_0 : \text{the sample follows distribution function } F(x) = x.$$

The KS test for this sample yields the KS statistic $D_N = 0.06127$ using four significant digits. We also compute the modified test statistic [11], obtaining $\tilde{D}_N = 1.9452$. Even at the 1% significance level (table [1]), we have $\tilde{D}_N > \alpha$. Therefore, $\tilde{D}_N$ is unusually large and we *reject* the null hypothesis. We have statistically significant evidence to believe that the generator does not output i.i.d. random samples from the uniform distribution on $(0,1)$.

We will now focus on the two empirical tests described in the previous section. We begin with an exercise where we compute the test statistic for the serial correlation test.

Chapter 2 Problem 8

Show that the serial correlation coefficient ρ is equal to -1 if $n = 2$, provided the denominator $\neq 0$.

For $n = 2$, the sample sequences U and V become:

$$U : u_1, u_2$$
$$V : u_2, u_1$$

Therefore, taking $n = 2$ in the equation for the serial correlation coefficient [12], we show

$$\begin{aligned}
\rho &= \frac{2\sum_{i=1}^{2} u_i u_{i+1} - (\sum_{i=1}^{2} u_i)^2}{2\sum_{i=1}^{2} u_i^2 - (\sum_{i=1}^{2} u_i)^2} \\
&= \frac{2(u_1u_2 + u_2u_3) - (u_1 + u_2)^2}{2(u_1^2 + u_2^2) - (u_1 + u_2)^2} \\
&= \frac{2(u_1u_2 + u_2u_1) - (u_1 + u_2)^2}{2(u_1^2 + u_2^2) - (u_1 + u_2)^2}, \text{ since } u_3 = u_{2+1} = u_1, \\
&= \frac{4u_1u_2 - (u_1^2 + 2u_1u_2 + u_2^2)}{2u_1^2 + 2u_2^2 - (u_1^2 + 2u_1u_2 + u_2^2)} \\
&= \frac{-u_1^2 + 2u_1u_2 - u_2^2}{u_1^2 - 2u_1u_2 + u_2^2} \\
&= \frac{-(u_1^2 - 2u_1u_2 + u_2^2)}{u_1^2 - 2u_1u_2 + u_2^2} \\
&= -1.
\end{aligned}$$

For two sample values, we only observe the correlation between (u_1, u_2) and (u_2, u_1). Since both pairs contain exactly the same values, we expect the two samples will be perfectly negatively correlated, and the result $\rho = -1$ confirms these observations.

Lastly, we construct an algorithm for the run-up test and use it to evaluate the quality of a chosen PRNG:

Chapter 2 Problem 12

Implement the run-up test and apply it to the MCG

$$x_n \equiv 8x_{n-1} \mod 67100963$$

Take the seed as 123456, and $n = 10,000$ in the run test. What are your conclusions?

We generate a sequence of $n = 10,000$ pseudo-random variables from the MCG, and find each run-up of length i in the sequence. The following output contains the count of run-ups of length $1, \ldots, 5$ or of length 6 or more:

```
1: 1988
2: 1835
3: 813
4: 314
5: 96
6: 27
```

Observing this output, it is evident that a majority of the sequence contains run-ups of length 1 or 2. We want to test these observed frequencies under the null hypothesis, which states

$$H_0 : \text{the sample is i.i.d. from U(0,1) (follows no specific trend).}$$

Computing the run-up statistic [15] and using the coefficients we defined for $k = 6$ [16], we obtain $r_u = 172.1$ for four significant digits. For large n, we know r_u has an approximate χ^2 distribution with 6 degrees of freedom. We interpret the test statistic by comparing r_u to the critical value for $\chi^2(6)$ [2]. The critical values for varying percentiles are:

```
90% critical value:10.64
95% critical value:12.59
99% critical value:16.81
```

Since r_u is significantly larger than the critical value at the 0.1% tail cutoff (or 99th percentile) then we *reject* the null hypothesis. Therefore, we must conclude that the sample from this generator does not behave like a i.i.d. $U(0, 1)$ random sequence.

Appendix

Complete Code

```
N1, N2 = 1000, 10000
seed = 123456
x0, x1 = 1, 1

# define coefficients for run-up test run-up test statistic
b = [1/6, 5/24, 11/120, 19/720, 29/5040, 1/840]
a = [
    4529.4   9044.9  13568   18091   22615   27892;
    9044.9   18097   27139   36187   45234   55789;
    13568    27139   40721   54281   67852   83685;
    18091    36187   54281   72414   90470   111580;
    22615    45234   67852   90470   113262  139476;
    27892    55789   83685   111580  139476  172860
]

# ---------------------
# Chapter 2 Problem 4
# ---------------------
function fibonacci_generator(N, x0, x1)
    sample = [x0, x1]
    for k in 1:(N-2)
        x2 = mod(x1 + x0, 2^31) # lagged Fibonacci generator
        push!(sample, x2) # collect each generated sample
        x0 = x1
        x1 = x2
    end
    return sample # return entire pseudorandom sequence
end

function ks_test(sample::Array)
    N=length(sample)
    order=sort(sample)
    dplus=0
    dminus=0
    # compute KS-statistic -> take maximum difference in empirical and theoretical cdf values
    for k in 1:N
        dplus=max(dplus,k/N-order[k])
    end
    for k in 1:N
        dminus=max(dminus,order[k]-(k-1)/N)
    end
    return max(dplus,dminus) # return the KS-statistic
end
```

```julia

# ---------------------
# Chapter 2 Problem 12
# ---------------------
function runup_test(N, seed)
    x0 = seed
    runups = Int[]
    run_length = 1
    for i in 0:N
        x1 = mod(8*x0, 67100963) # multiplicative congruential generator
        if x1 > x0
            run_length += 1 # length of run i
        else
            push!(runups, run_length)
            run_length = 1 # reset count when run-up ends (next value < previous value)
        end
        x0 = x1
    end
    push!(runups, run_length)
    return runups # return all run-ups in sequence
end

function runup_statistic(runups, a, b, N)
    # count number of run-ups of length 1-5 or length 6 or more
    u = [
        count(==(1), runups),
        count(==(2), runups),
        count(==(3), runups),
        count(==(4), runups),
        count(==(5), runups),
        count(x -> x >= 6, runups)
    ]
    r = 0.0
    for j in 1:6
        for i in 1:6
            r += (u[i] - N*b[i]) * (u[j] - N*b[j]) * a[i, j] # compute run-up statistic w/ pre-defined coefficients
        end
    end
    return r / N
end
```

```r
# ---------------------
# Test Results
# ---------------------

# Apply the KS-test to the Fibonacci generator for N=1000 samples w/ seeds x0=x1=1
sample = fibonacci_generator(N1, x0, x1)
ks_stat = ks_test(sample)
println(ks_stat)
modified_ks_stat = ks_stat * (sqrt(N1) + 0.12 + (0.11/sqrt(N1))) # compute modified test statistic for large sample size
println(modified_ks_stat)

# Perform the run-up test for N=10000 samples w/ seed=123456
runs = runup_test(N2,seed)
run_stat = runup_statistic(runs, a, b, N2)
println(run_stat)
# calculate 90%,95%,99% critical values for chi-squared disttribution with 6 degrees of freedom
d = Chisq(6)
critical_value_90 = quantile(d, 0.90)
critical_value_95 = quantile(d, 0.95)
critical_value_99 = quantile(d, 0.99)
println("90% critical value:", critical_value_90)
println("95% critical value:", critical_value_95)
println("99% critical value:", critical_value_99)
```

References

[1] G. Okten, *Monte Carlo and quasi-Monte Carlo Methods in Computational Finance*, Draft, 2026.

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[3] "Kolmogorov-Smirnov Test," *Wikipedia, The Free Encyclopedia*. Available at: https://en.wikipedia.org/wiki/KolmogorovSmirnov_test (Accessed: February 23, 2026).